Subject: Mathematics | **WEEKLY CHAPTER TEST – 03 (PROBABILITY)** | **Time:** 45 Minutes

Class: Class X (CBSE) | **Chapter 14: Probability** | **Max. Marks: 20**

SECTION A: OBJECTIVE & MCQs (1 Mark Each) [5 Marks]

Question 1 [1 Mark]

Three unbiased coins are tossed together. The probability of getting exactly two tails is:

(a) $\frac{1}{8}$ (b) $\frac{3}{8}$

(c) $\frac{1}{2}$ (d) $\frac{5}{8}$

Working & Calculation:

- When three fair coins are tossed, the complete sample space S is:

$$S = \{\text{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT}\} \implies n(S) = 2^3 = 8$$

- Let E be the event of getting **exactly two tails**:

$$E = \{\text{HTT, THT, TTH}\} \implies n(E) = 3$$

- By the classical definition of probability:

$$P(\text{exactly two tails}) = \frac{n(E)}{n(S)} = \frac{3}{8}$$

CORRECT ANSWER: OPTION (b) $\longrightarrow \frac{3}{8}$

Question 2 [1 Mark]

Two dice are thrown simultaneously. The probability of getting a doublet is:

(a) $\frac{1}{36}$ (b) $\frac{1}{6}$

(c) $\frac{5}{6}$ (d) $\frac{1}{12}$

Working & Calculation:

- Total possible elementary outcomes when two dice are rolled: $n(S) = 6 \times 6 = 36$.
- A **doublet** occurs when both dice display the identical face value:

$$E = \{(1,1), (2,2), (3,3), (4,4), (5,5), (6,6)\} \implies n(E) = 6$$

- Therefore:

$$P(\text{a doublet}) = \frac{n(E)}{n(S)} = \frac{6}{36} = \frac{1}{6}$$

CORRECT ANSWER: OPTION (b) $\longrightarrow \frac{1}{6}$

Question 3 [1 Mark]

A bag contains 5 red balls and n green balls. If the probability of drawing a green ball is three times that of drawing a red ball, then the value of n is:

(a) 10 **(b) 15**

(c) 20 **(d) 18**

Working & Calculation:

- Total balls in bag: $n(S) = 5 + n$.
- Given $P(\text{Green}) = 3 \times P(\text{Red})$:

$$\frac{n}{5+n} = 3 \times \frac{5}{5+n} \implies \frac{n}{5+n} = \frac{15}{5+n} \implies n = 15$$

CORRECT ANSWER: OPTION (b) $\longrightarrow$ 15

Question 4 [1 Mark]

The probability that a leap year chosen at random has 53 Sundays is:

(a) $\frac{1}{7}$ **(b) $\frac{2}{7}$**

(c) $\frac{3}{7}$ **(d) $\frac{5}{7}$**

Working & Calculation:

- A leap year has 366 days = 52 weeks + 2 extra days.
- The 2 consecutive extra days have 7 equally likely outcomes:

$$S = \{(\text{Sun, Mon}), (\text{Mon, Tue}), (\text{Tue, Wed}), (\text{Wed, Thu}), (\text{Thu, Fri}), (\text{Fri, Sat}), (\text{Sat, Sun})\}$$

- Pairs containing Sunday: $E = \{(\text{Sun, Mon}), (\text{Sat, Sun})\} \implies n(E) = 2$:

$$P(53 \text{ Sundays}) = \frac{n(E)}{n(S)} = \frac{2}{7}$$

CORRECT ANSWER: OPTION (b) $\longrightarrow$ $\frac{2}{7}$

Question 5 (Assertion–Reason) [1 Mark]

Assertion (A): In a single throw of a die, the probability of getting an even prime number is $\frac{1}{6}$.
Reason (R): The only even prime number on a standard die is 2, which is 1 favourable outcome out of 6.

Options:

(a) Both (A) and (R) are true, and (R) is the correct explanation of (A).
(b) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
(c) (A) is true, but (R) is false.
(d) (A) is false, but (R) is true.

Step-by-Step Logical Analysis:

1. **Assertion (A):** On a die $\{1, 2, 3, 4, 5, 6\}$, primes are $\{2, 3, 5\}$. The only even prime is 2. Thus favourable outcome is 1 $\implies P = \frac{1}{6}$. $\implies$ **(A) is TRUE.**
2. **Reason (R):** 2 is the unique even prime on a standard die (1 outcome out of 6). $\implies$ **(R) is TRUE.**
3. **Explanation Check:** (R) states the exact mathematical rationale for (A). Hence, (R) correctly explains (A).

CORRECT ANSWER: OPTION (a) $\longrightarrow$ Both (A) and (R) are true, and (R) is the correct explanation of (A).

SECTION B: VERY SHORT ANSWER QUESTIONS (2 Marks Each) [2 Marks]

Question 6 [2 Marks]

One card is drawn at random from a well-shuffled pack of 52 playing cards. Find the probability of getting: (i) an ace (ii) a black king.

Sample Space: Total number of playing cards in a standard deck: $n(S) = 52$.

(i) Probability of getting an ace [1 Mark]

A standard deck of 52 cards contains exactly 4 aces (one each in Spades, Hearts, Diamonds, and Clubs):

$$n(\text{aces}) = 4 \implies P(\text{an ace}) = \frac{n(\text{aces})}{n(S)} = \frac{4}{52} = \frac{1}{13}$$

Answer (i): $\mathbf{P(\text{an ace}) = \frac{1}{13}}$

(ii) Probability of getting a black king [1 Mark]

The black suits are Spades and Clubs. Each suit has 1 king, giving a total of 2 black kings:

$$n(\text{black kings}) = 2 \implies P(\text{a black king}) = \frac{n(\text{black kings})}{n(S)} = \frac{2}{52} = \frac{1}{26}$$

Answer (ii): $\mathbf{P(\text{a black king}) = \frac{1}{26}}$

SECTION C: SHORT ANSWER QUESTIONS (3 Marks Each) [9 Marks]

Question 7 [3 Marks]

Two dice are thrown together. Find the probability that:
(i) the product of the numbers is 12 (ii) the product of the numbers is odd (iii) the product of the numbers is even.

Sample Space: When two dice are thrown together, the total number of elementary outcomes is $n(S) = 6 \times 6 = 36$.

(i) Product of the numbers is 12 [1 Mark]

Outcomes (a, b) with $a \times b = 12$ are $E_1 = \{(2,6), (3,4), (4,3), (6,2)\}$:

$$n(E_1) = 4 \implies P(\text{product is 12}) = \frac{n(E_1)}{n(S)} = \frac{4}{36} = \frac{1}{9}$$

Answer (i): $\mathbf{P(\text{product is 12}) = \frac{1}{9}}$

(ii) Product of the numbers is odd [1 Mark]

The product $a \times b$ is odd if and only if both a and b are odd numbers from $\{1, 3, 5\}$ ($3 \times 3 = 9$ outcomes):

$$E_2 = \{(1,1), (1,3), (1,5), (3,1), (3,3), (3,5), (5,1), (5,3), (5,5)\} \implies n(E_2) = 9$$

$$P(\text{product is odd}) = \frac{n(E_2)}{n(S)} = \frac{9}{36} = \frac{1}{4}$$

Answer (ii): $\mathbf{P(\text{product is odd}) = \frac{1}{4}}$

(iii) Product of the numbers is even [1 Mark]

The event of the product being even is the exact complement of the product being odd:

$$P(\text{product is even}) = 1 - P(\text{product is odd}) = 1 - \frac{1}{4} = \frac{3}{4} \quad \left[\text{or } \frac{36-9}{36} = \frac{27}{36} = \frac{3}{4}\right]$$

Answer (iii): $\mathbf{P(\text{product is even}) = \frac{3}{4}}$

Question 8 [3 Marks]

A bag contains 15 balls of which x are blue and the remaining are red.
(i) If a ball is drawn at random, write the expression for the probability that it is a blue ball.
(ii) If 5 more blue balls are added to the bag, the probability of drawing a blue ball becomes double that in part (i). Find the value of x.

Given Data: Total initial balls $= 15$, blue balls $= x$, red balls $= 15 - x$.

(i) Expression for the initial probability of drawing a blue ball [1 Mark]

Since there are x blue balls out of a total of 15 balls:

$$P(\text{blue ball initially}) = \frac{x}{15}$$

Answer (i): $\mathbf{P(blue) = \frac{x}{15}}$

(ii) Finding the value of x after adding 5 blue balls [2 Marks]

When 5 more blue balls are added to the bag:

- New number of blue balls $= x + 5$.
- New total number of balls in the bag $= 15 + 5 = 20$.
- New probability of drawing a blue ball: $P'(\text{blue}) = \frac{x+5}{20}$.

Given that the new probability is double the initial probability:

$$P'(\text{blue}) = 2 \times P(\text{blue}) \implies \frac{x+5}{20} = 2 \times \frac{x}{15} \implies \frac{x+5}{20} = \frac{2x}{15}$$

Cross-multiplying to solve for x:

$$15(x+5) = 20(2x) \implies 15x + 75 = 40x \implies 40x - 15x = 75 \implies 25x = 75 \implies x = 3$$

Answer (ii): $\mathbf{x = 3}$

Question 9 [3 Marks]

Cards numbered from 1 to 49 are placed in a box and mixed thoroughly. A card is drawn at random from the box. Find the probability that the number on the card is:
(i) an odd number (ii) a perfect square (iii) a multiple of 7.

Sample Space: Cards are numbered $1, 2, 3, \ldots, 49 \implies n(S) = 49$.

(i) An odd number [1 Mark]

The odd numbers in the set $\{1, 2, \ldots, 49\}$ form an arithmetic sequence $1, 3, 5, \ldots, 49$:

$$\text{Count } n(E_1) = \frac{49-1}{2} + 1 = 25 \implies P(\text{odd number}) = \frac{n(E_1)}{n(S)} = \frac{25}{49}$$

Answer (i): $\mathbf{P(odd\ number) = \frac{25}{49}}$

(ii) A perfect square [1 Mark]

The perfect squares between 1 and 49 are $E_2 = \{1^2, 2^2, 3^2, 4^2, 5^2, 6^2, 7^2\} = \{1, 4, 9, 16, 25, 36, 49\}$:

$$n(E_2) = 7 \implies P(\text{perfect square}) = \frac{n(E_2)}{n(S)} = \frac{7}{49} = \frac{1}{7}$$

Answer (ii): $\mathbf{P(perfect\ square) = \frac{1}{7}}$

(iii) A multiple of 7 [1 Mark]

The multiples of 7 in the range 1 to 49 are $E_3 = \{7, 14, 21, 28, 35, 42, 49\}$:

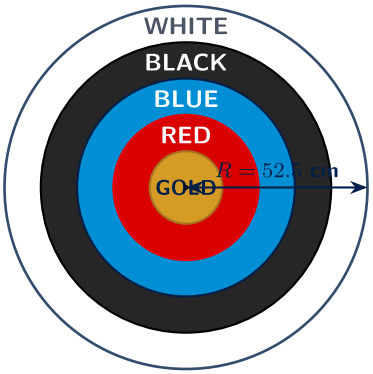
$$n(E_3) = 7 \implies P(\text{multiple of 7}) = \frac{n(E_3)}{n(S)} = \frac{7}{49} = \frac{1}{7}$$

Answer (iii): $\mathbf{P(multiple\ of\ 7) = \frac{1}{7}}$

SECTION D: APPLIED CASE STUDY (4 Marks) [4 Marks]

Question 10 (Case Study Analysis) [4 Marks]

In a national archery championship, an Olympic target is designed with five concentric scoring circles marked from the center outwards as Gold, Red, Blue, Black, and White. The diameter of the central Gold region is 21 cm and each of the four surrounding concentric bands has a uniform width of 10.5 cm. An archer releases an arrow that hits the target at random.



Geometric Parameters:

- Diameter of Gold region $= 21$ cm $\implies$ Radius $r_1 = \dfrac{21}{2} = 10.5$ cm.
- Each of the four outer concentric bands has a uniform width of $w = 10.5$ cm.
- Total target radius $R = r_5 = 10.5 + 4(10.5) = 5 \times 10.5 = 52.5$ cm.
- Total target area: $A_{\text{total}} = \pi R^2 = \pi(5 \times 10.5)^2 = 25\pi(10.5)^2$.

(i) Radius of the central Gold region and total radius of the entire board [1 Mark]

Radius of Gold region: $r_{\text{Gold}} = \dfrac{21}{2} = 10.5$ cm. Total radius: $R = 10.5 + 4(10.5) = 52.5$ cm.

Answer (i): $\mathbf{r_{Gold} = 10.5\ cm}$, $\mathbf{R_{total} = 52.5\ cm}$

(ii) Probability that the arrow hits the Gold region [1 Mark]

Area of Gold $= \pi r_1^2 = \pi(10.5)^2$. By geometric probability:

$$P(\text{Gold}) = \frac{\text{Area of Gold}}{\text{Total Area}} = \frac{\pi(10.5)^2}{25\pi(10.5)^2} = \frac{1}{25}$$

Answer (ii): $\mathbf{P(Gold) = \dfrac{1}{25}}$

(iii) Probability that the arrow hits the Red ring [1 Mark]

The Red region is the annular ring between $r_1 = 10.5$ cm and $r_2 = 10.5 + 10.5 = 21$ cm:

$$\text{Area of Red} = \pi(r_2^2 - r_1^2) = \pi\left[(2 \times 10.5)^2 - (10.5)^2\right] = \pi(10.5)^2[4 - 1] = 3\pi(10.5)^2$$

$$P(\text{Red ring}) = \frac{\text{Area of Red}}{\text{Total Area}} = \frac{3\pi(10.5)^2}{25\pi(10.5)^2} = \frac{3}{25}$$

Answer (iii): $\mathbf{P(Red\ ring) = \dfrac{3}{25}}$

(iv) Probability that the arrow does NOT hit the Gold region [1 Mark]

This is the complementary event of hitting the Gold region:

$$P(\text{not Gold}) = 1 - P(\text{Gold}) = 1 - \frac{1}{25} = \frac{24}{25}$$

Answer (iv): $\mathbf{P(not\ Gold) = \dfrac{24}{25}}$

— END OF WORKED SOLUTIONS (TEST–03) · MATHS MANTHAN TUITION CLASSES —